

DUSTstreaks: A Stratospheric Star-Tracker Dataset and Detection Benchmark for Space Situational Awareness

Acquisition

RSONAR 2022
stratospheric
balloon flight
**Dual Use Star
Tracker
(DUST)**

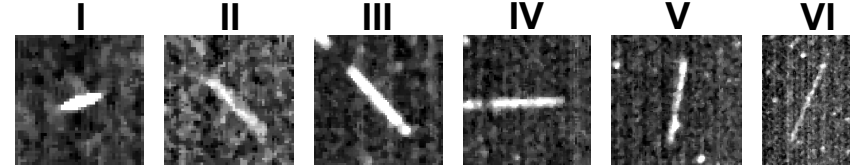
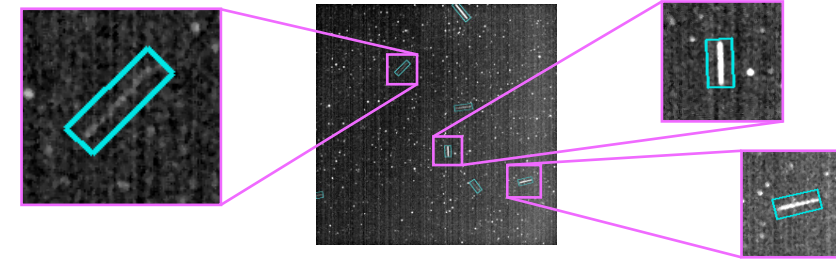


Preprocessing

Stack 27
exposures
Bias & flat
correction
Denoise &
stretch



DUSTstreaks Dataset



Morphological groups

Detection Benchmark

Best F_1 scores by detector

Faster R-CNN 80.2%

RetinaNet 78.8%

FCOS 76.8%

YOLOX-tiny 76.5%

Public Release



625 images



1574 annotations



203 unique transits



model weights